HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · PROFESSIONAL SERVICES · FIELD RESEARCH

NATIONAL SKILLS STUDY · SIX STATES · 6,000 FIELD SIGNATURES

Field Data Digitisation

Thousands of respondent signatures sat on creased, noisy A4 scans. An Excel VBA tool now captures each one, files it under the right respondent's name and undoes a bad capture in one click.

~~200~~ → 8 hours to extract the full signature set

8 hours
Turnaround for the signature set
▼ from 200 hours

96 %
Reduction in turnaround time
▼ 200 hours down to 8

5 seconds
Time per signature
▼ from about 2 minutes

0 files
Files saved under the wrong name
auto-named from the respondent list

Rings: 8 of the old 200 hours; 96% of turnaround removed; 5 of the old 120 seconds per signature; every file named correctly. Figures indicative; validated metrics available on request.

01 200 hours became 8

Each square is one hour of analyst time · indicative

BEFORE · BY HAND — 200 analyst hours: crop, paste, rename, save, check

AFTER · WITH THE TOOL — 8 hours with the tool · 192 hours saved

8 h the person still frames each signature; naming, saving and tracking take seconds

200 h	8 h	192 h	96%
by hand	with the tool	saved	less turnaround

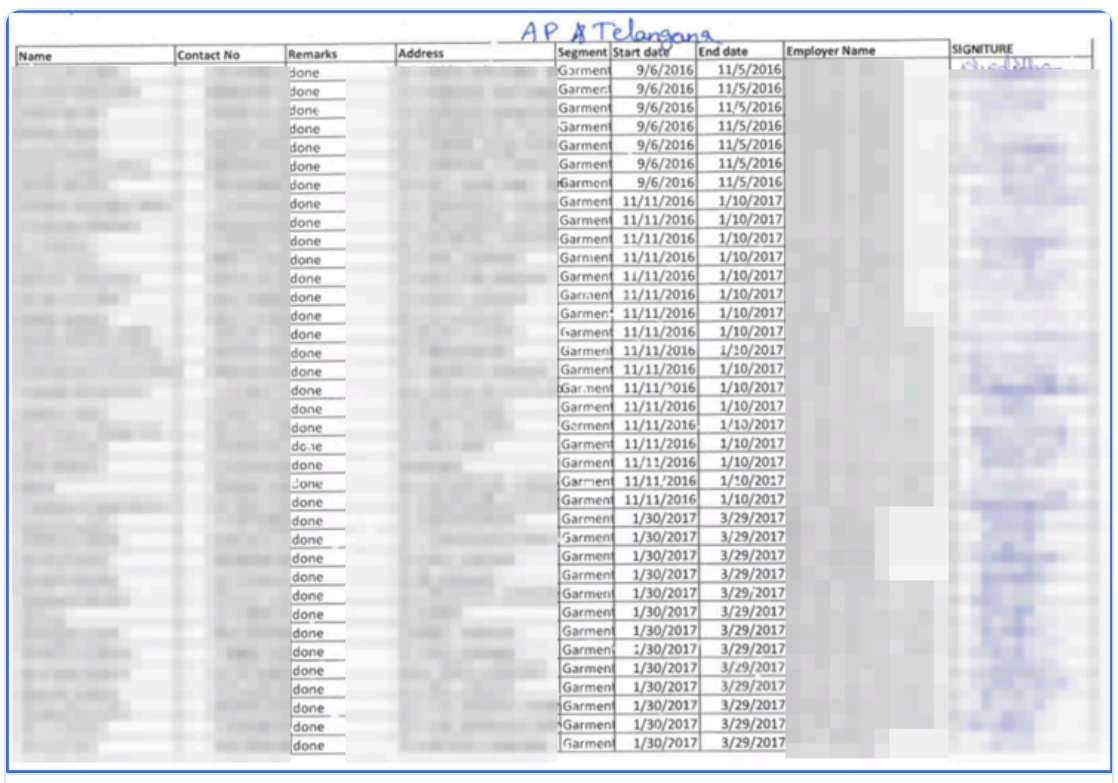
THE PROBLEM
Automation alone failed
Folds, creases and uneven scans defeated automatic signature detection on many sheets.

THE SPLIT
Person frames, macro files
A person draws one box around each signature. Naming, saving, highlighting and moving on happen in the macro.

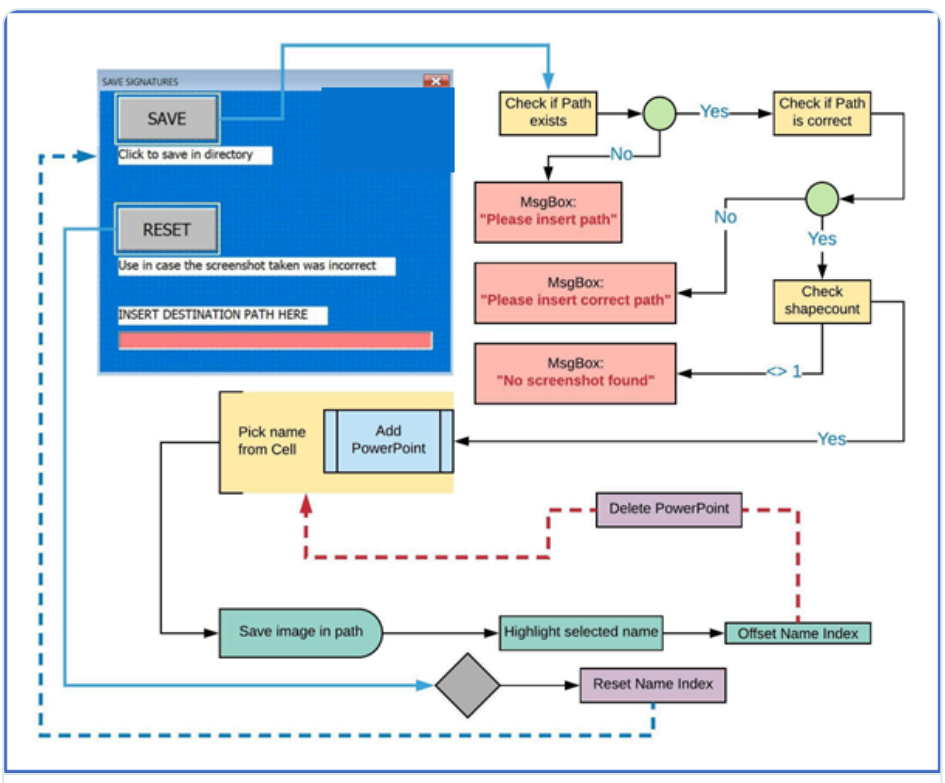
THE RESULT
A named library, ready to validate
Every file matched the respondent list exactly, so contacts could be checked straight away.

02 The tool, as delivered

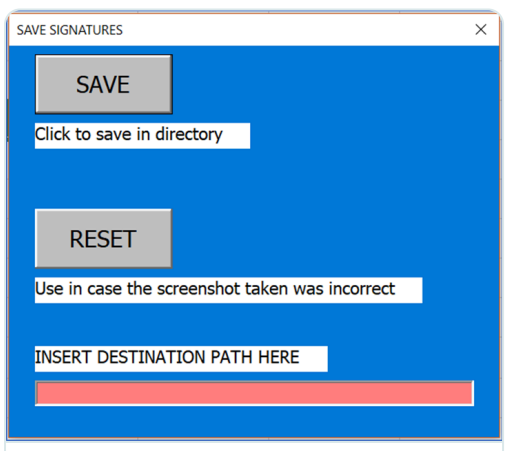
Original screens · personal details pixelated



1 **A field sheet, AP & Telangana**: one row per respondent, dozens per sheet, with the scan noise and uneven lines the tool had to cope with



2 **Macro flow**: SAVE checks the path exists, is correct and holds exactly one capture, then names, saves, highlights and moves on; RESET deletes and steps back



3 **SAVE SIGNATURES form**: two buttons and a path box

ONE CAPTURE, START TO FINISH

1 **Snip** — The person frames one signature beside the respondent list.

2 **SAVE** — Guard checks run; the name is picked from the cell, never typed.

3 **Export** — The capture is saved as an image named after the respondent.

4 **Highlight** — The row turns highlighted and the name index moves to the next row.

5 **RESET** — Only if the capture was wrong: deletes it and steps the index back.

Message boxes: "Please insert path" · "Please insert correct path" · "No screenshot found"

03 Four pain points, four fixes

Before → after

BEFORE
Crop by hand
Crop in an image tool, paste, adjust, for every signature.

SCREEN CAPTURE

AFTER
Snip beside the list
The scan sits next to the respondent list in Excel; one snip per signature.

BEFORE
Names typed by hand
Most errors were the right image under the wrong respondent.

RESPONDENT LIST

AFTER
Never typed
Names come from the list, in order, so files match for validation.

BEFORE
Save As, every time
Browse to the folder for each file; progress tracked on paper.

SAVE MACRO

AFTER
Auto-saved, auto-tracked
Exported to the assigned folder; each saved name highlighted in the sheet.

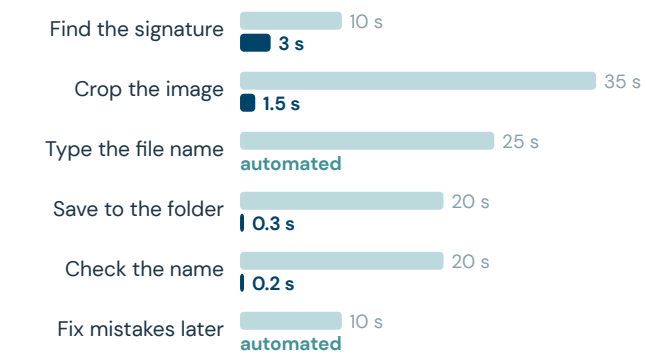
BEFORE
Mistakes found late
Wrong captures surfaced only during validation.

RESET BUTTON

AFTER
Undo in one click
RESET deletes the last capture and steps the name index back.

04 Where 2 minutes went

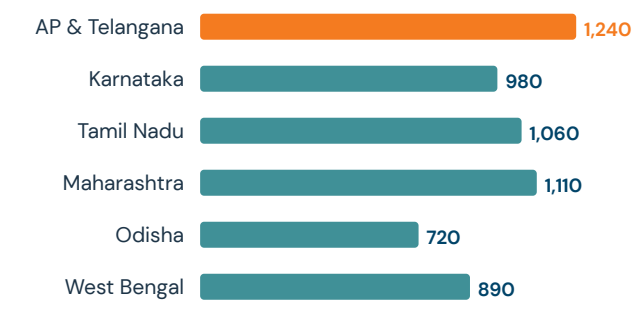
Seconds per signature by step, before (light) and with the tool (navy)



120 s → 5 s per signature

05 Same tool, every state

Signatures per state batch · state split illustrative



6,000	8.0 h	200 h
signatures	analyst hours, tool	same work by hand

Four analysts in parallel: about 2 hours elapsed

06 How HMC ran it

2 days	**Discover**	Timed the manual way: about 2 minutes a signature; errors mostly in naming.
2 days	**Design**	Judgement for a person, routine for the macro. Flow chart and form layout.
1 week	**Build**	Macro-enabled workbook with the SAVE SIGNATURES form.
1 day	**Adopt**	Ten-minute training and a one-page guide; productive the same day.
∞ all states	**Scale**	Batches split by state and run in parallel, one folder per state and employer.

What any research team can borrow

01 Automate around the judgement
When scans are too messy for full automation, keep the person on the hard bit and automate every step around it.

02 Never type a name twice
Taking names from the respondent list removed the most common error: the right image under the wrong name.

03 Make undo one click
RESET made mistakes cheap to fix in the moment, instead of costly to find during validation.

Built with **Excel VBA** · **VBA UserForm** · **Screen capture** · **Chart export** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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